Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2024

Course: B.Tech
Code: 155501

Subject: IoT Architecture & Design

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer from the following (Any seven question only):*** **[2 x 7 = 14]**

(a) IoT stands for
- (i) Internet of Tools
- (ii) Internet of Things
- (iii) Interoffice Technology
- (iv) Intelligent of Things

(b) Which of the following is a hardware component of IoT?
- (i) Sensor
- (ii) Cloud
- (iii) Software
- (iv) Protocol

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(c) Raspberry Pi is used for
- (i) Data entry
- (ii) Embedded programming
- (iii) Image editing
- (iv) None

(d) Which of the following is not a wireless protocol?
- (i) ZigBee
- (ii) Bluetooth
- (iii) CoAP
- (iv) Wi-Fi

(e) Which of the following is a type of actuator?
- (i) Temperature sensor
- (ii) Motor
- (iii) Camera
- (iv) Thermistor

(f) Digital sensors produce
- (i) Continuous signals
- (ii) Analog outputs
- (iii) Binary signals
- (iv) Optical signals

(g) Which of the following is not a component of IoT architecture?
- (i) Sensors
- (ii) Actuators
- (iii) APIs
- (iv) Database

(h) How many main layers are consists of IoT architecture?
- (i) 2
- (ii) 3
- (iii) 4
- (iv) 5

(i) REST stands for
- (i) Ready Encoding Set Transfer
- (ii) Representational State Transfer
- (iii) Real-time Service Terminal
- (iv) Remote Encoding State Transfer

(j) What is a major security concern in IoT?
- (i) High bandwidth
- (ii) IP conflict
- (iii) Data privacy
- (iv) Cloud cost

Q.2 (a) Define IoT and explain any four applications of IoT. **[7]**
(b) Discuss the role of cloud computing in IoT systems. **[7]**

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Q.3	(a)	What is MQTT? Explain its role in IoT communication.	[7]
	(b)	Write down the Comparisons between Arduino and Raspberry Pi with respect to IoT applications.	[7]
Q.4	(a)	Differentiate between sensor and transducer with suitable examples.	[7]
	(b)	Describe the evolution of IoT sensors from first generation to integrated systems.	[7]
Q.5	(a)	Describe the importance of deployment and operational view in IoT systems.	[7]
	(b)	Explain the various components of IoT architecture.	[7]
Q.6	(a)	Define REST API with a suitable IoT example.	[7]
	(b)	Define URI. How is it used to identify web-based resources in IoT?	[7]
Q.7	(a)	Describe E-health systems with an example.	[7]
	(b)	Discuss the privacy and security issues in IoT with any solutions.	[7]
Q.8	(a)	Explain the conceptual framework and layers of an IoT architecture.	[7]
	(b)	Describe the software APIs used in IoT.	[7]
Q.9	(a)	Write a short note on polytronics systems and sensor swarms.	[7]
	(b)	Write a short note on the functional view of IoT architecture.	[7]

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